

Sec-A

1. What is known as strobe signal?

Ans A signal used to time reading that is output together with the data. A strobe signal is required to connect to devices for high-speed input or high-speed response such as microcomputers boards.

2. What is data bus?

Ans A data bus is a communication system within a computer or between computers that allows for the transfer of data between various components such as the CPU, memory.

3. What is memory-mapped?

Ans Memory mapped refers to a technique used in computer system where hardware devices, such as input/output (I/O) devices, are accessed using the same address space as the computer's memory.

4. What is handshaking?

Ans Handshaking refers to the process by which two devices establish and synchronize communication with each other. It involves a series of predefined signals or messages exchanged between the devices to negotiate parameters such as data transmission rate, protocol compatibility and readiness to send or receive data.

5. Explain interface.

Ans An interface defines how different hardware components communicate with each other. Interface ensure compatibility and efficient communication within the system.

Sec-B (3x7=21)

1. Explain the following terms: Bus grant and Bus request.

Ans in computer arch. a "bus grant" and "bus request" are terms related to bus arbitration, which is the process of determining which device on a shared bus can access it at a given time.

(i) Bus Request (BR, REQ or BRQ): When a device wants to communicate over a shared bus, it sends a bus request signal to the bus controller or arbiter. This signal indicates the device's desire to use the bus for data transfer.

(ii) Bus Grant (BG or BGRT): The bus arbiter, upon receiving multiple bus requests, decides which device gets permission to use the bus next. It grants access to the bus by sending a bus grant signal to the selected device, indicating that it can proceed with data transfer.

2. What is valid data in terms of address bus and data bus.

Ans Valid data for an address bus and a data bus depends on their respective functionalities.

(i) ~~Valid data on~~ Address bus: This bus carries memory locations a device wants to access. Valid data here refers to a binary number that falls within the addressable range of the system. This range is determined by the width (number of bits) of the address bus.

For example: - A 16-bit address bus can represent

addresses from 0 (0000...00) to $2^{16}-1$ (1111...11). Any binary number outside this range would not be a valid address.

(ii) Data bus: This bus carries the actual data being transferred between components. Valid data here depends on the data type and the width of the data bus.

- Data Type: The data could be instructions, no., text or any digital information the system works with. The format determines how it's represented in binary.

- Data Bus width: The data bus width (no. of bits) defines how much data can be transferred at once.

3. Explain polled and Interrupt-driven I/O with suitable structure.

Ans Polled I/O also known as programmed I/O or software I/O, involves the CPU periodically checking the status of external devices to see if they require attention or data transfer. The CPU initiates and controls the data transfer bet. the device and memory. In a polled interrupt scheme, each device is tested until the device requesting the interrupt is identified. The CPU then diverts the program to the interrupt Service Routine (ISR) written for specific device. This method is used when multiple devices are connected to the system.

- Hardware Polling Scheme:
 - Involves checking the status of devices using a daisy-chaining method.

→ The CPU identifies the interrupting device through the data bus.

- Software Polling Scheme:

→ involve checking the status of devices in software one after the other.

→ Devices are prioritized, with the highest priority device being serviced first.

- Interrupt-driven I/O:

Interrupt-driven I/O is a method where the CPU does not need to manually check the status of devices. Instead when an I/O request is available, the CPU is immediately notified using interrupts, and the request is handled through an Interrupt Service Routine (ISR). This method is more efficient as the CPU is not continuously checking for device status.

- Key points:

→ Interrupts immediately notify the CPU of I/O requests.

→ Interrupt-driven I/O is a hardware mechanism.

→ Devices are serviced by an interrupt handler.

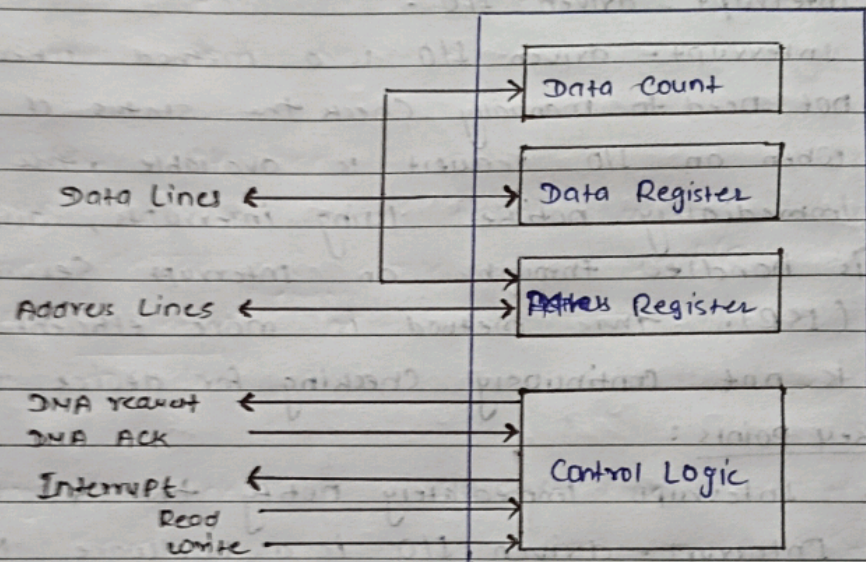
- Sec-C (3x11=33)

1. Explain in details about the DMA Controller.

Ans DMA stands for Direct Memory Access. DMA Controller is a hardware device that allows I/O devices to directly access memory with less participation of the processor. DMA Controller needs the same old circuitry of an interface to communication with the CPU and input/output devices.

Direct Access Memory access uses hardware for accessing the memory, that hardware is called a DMA controller. It has the work of transferring the data between input output device and main memory with very less interaction with the processor. The direct memory access controller is a control unit, which has the work of transferring data.

• DMA controller Diagram in Computer Architecture :



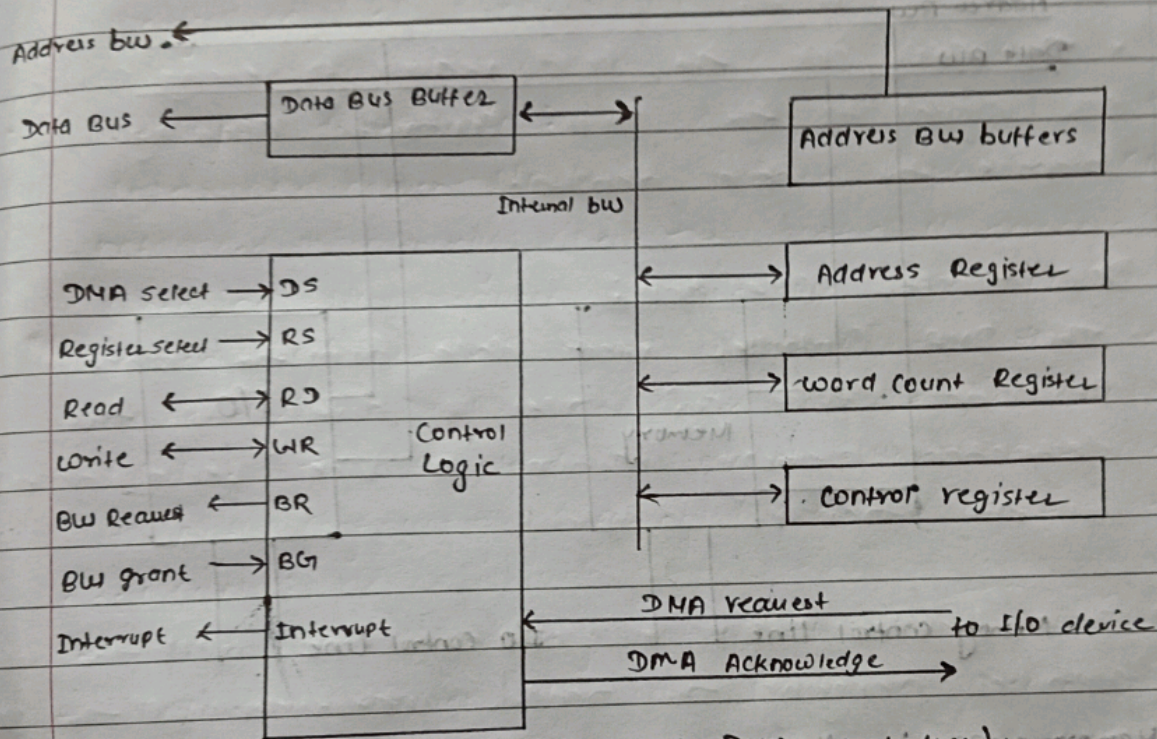
(Block diagram of DMA controller)

• Types of DMA :

There are many types of DMA.

- Single-Ended DMA
- Dual-Ended DMA
- Arbitrated-Ended DMA
- Interleaved DMA

- Working of DMA Controller:
 - The DMA Controller registers have three registers as follows:
 - Address register - It contains the address to specify the desired location in memory.
 - Word count register - It contains the number of words to be transferred.
 - Control register - It specifies the transfer mode.



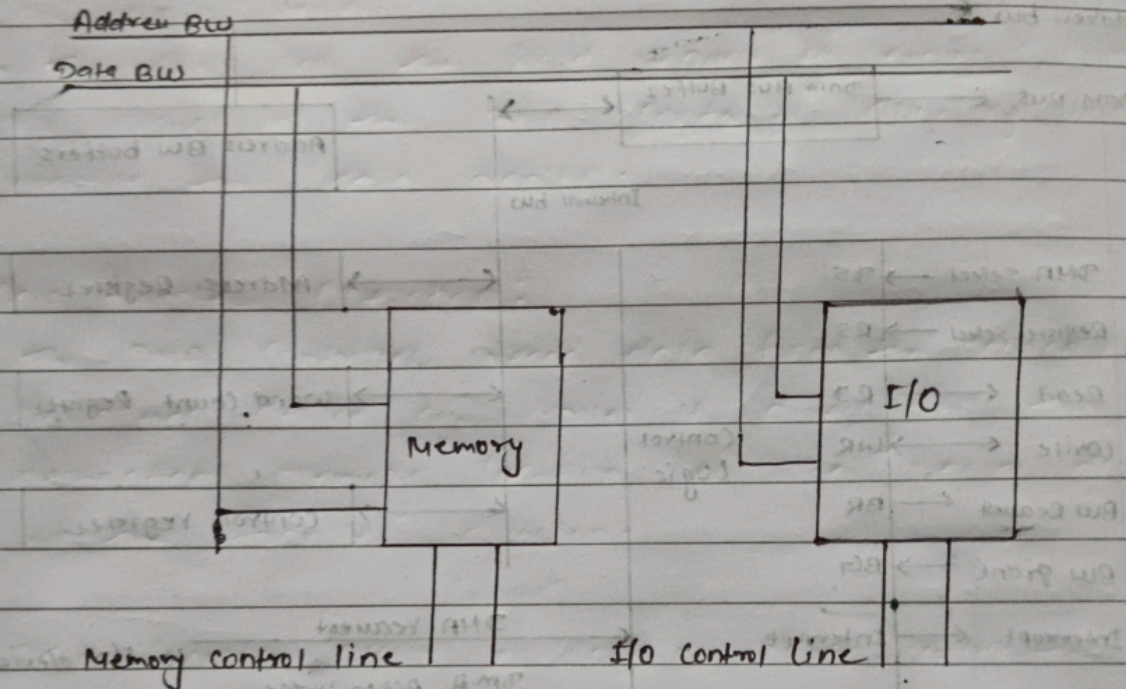
(Working diagram of DMA controller)

- Explanation -
 - The CPU initializes the DMA by sending the given information through the data bus.
 - The starting address of the memory block where the data is available (to read) or where data are to be stored (to write).
 - It also sends word count which is the number of words in the memory block to be read or written.
 - Control to define the mode of transfer such as read or write.
 - A Control to begin the DMA transfer.

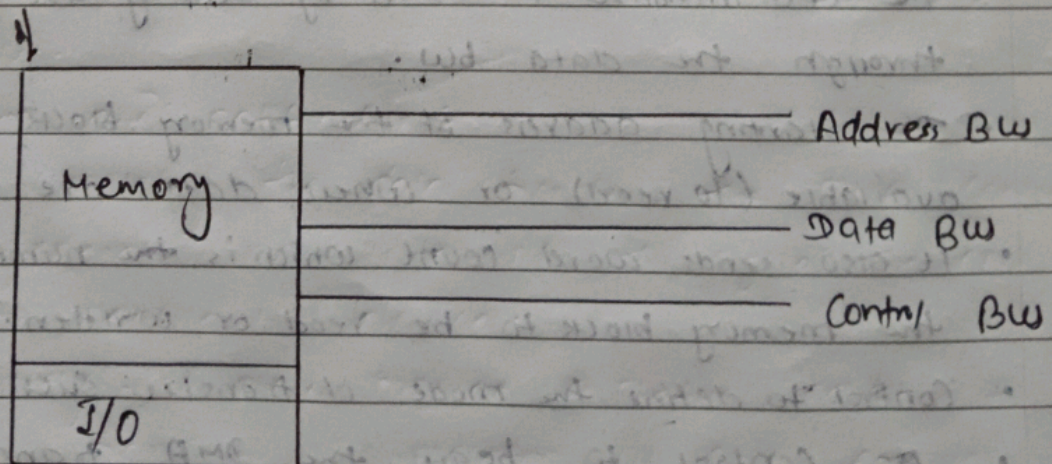
2. Write difference betn. isolated and memory mapped.

Ans. Isolated and memory mapped are two different approaches to input/output (I/O) operations in computer systems.

(i) Isolated I/O: In isolated I/O, special instructions distinct from regular memory operations are used to communicate with peripheral devices.



(ii) Memory mapped: Memory mapped I/O treats peripheral devices as if they are memory locations.



Isolated I/O	Memory Mapped I/O
• Memory and I/O have separate address space.	• Both have same address space.
• All address can be used by the memory.	• Due to addition of I/O addressable memory become less for memory.
• Separate Instruction control read and write operation in I/O and memory.	• Some instructions can control both I/O and memory.
• More efficient due to separate buses.	• Lesser efficient.
• Larger in size due to more buses.	• Smaller in size.
• It is complex due to separate logic is used to control both.	• Simpler logic is used as I/O is also treated as memory only.

3. Explain the DMA data transfer.

Ans DMA (direct memory access) data transfer is a method used in computer systems to transfer data between peripherals and memory without involving the CPU directly. Here's an explanation of how DMA data transfer works:

(i) Initialization:

Before initiating DMA data transfer, the CPU initializes the DMA controller by programming its registers with necessary parameters such as source and destination addresses, transfer size, and transfer mode.

(ii) Request and Acknowledge:

When a peripheral device requires data transfer, it sends a DMA request signal to the DMA controller.

The DMA Controller acknowledges the request and selects an available DMA channel to handle the transfer.

(iii) CPU Handover:

- Upon receiving the hold request, the CPU relinquishes control over the buses by tripping the data, address and control buses.
- The CPU ack the hold request by sending a hold ack signal to the DMA Controller, allowing it to manage the data transfer.

(iv) Data transfer Operation:

- The DMA Controller reads data from the source peripheral device and writes it directly to the destination memory location, or vice versa, depending on the transfer direction.
- Data transfer occurs in block or burst, with the DMA controller managing the transfer independently of the CPU.
- The CPU is typically free to perform other tasks while the DMA Controller handles the data transfer.

(v) Completion and notification:

- Upon completing the data transfer, the DMA Controller may generate an interrupt to signal the CPU or the peripheral device.
- Alternatively, it may provide a status indication to indicate transfer completion or any errors encountered during the transfer.

(vi) Resuming Execution:

- After handling the DMA transfer, the CPU resumes its execution, potentially processing the transferred data or performing other tasks.

DMA data transfer enables efficient and high-speed data movement between peripherals and memory by allowing the DMA controller to handle data transfer independently of the CPU, resulting in reduced CPU overhead and improved system performance.
